

Chapter 3: Convexity Fundamentals

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Convex Sets

We begin discussing convexity by defining what a convex set is.

Definition (Convex Set). *A set $C \subseteq \mathbb{R}^n$ is called convex if for any $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in [0, 1]$ the point $\lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$ belongs to C .*

In other words, the above definition is equivalent to saying that for any $\mathbf{x}, \mathbf{y} \in C$, the line segment $[\mathbf{x}, \mathbf{y}]$ is also in C .

Very Important Convex Sets

- A line in $\mathbb{R}^n$ is a set of the form

$$L = \{\mathbf{z} + t\mathbf{d} : t \in \mathbb{R}\}.$$

where $\mathbf{z}, \mathbf{d} \in \mathbb{R}^n$ and $\mathbf{d} \neq \mathbf{0}$.

- $[\mathbf{x}, \mathbf{y}]$, $(\mathbf{x}, \mathbf{y})$ for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ ($\mathbf{x} \neq \mathbf{y}$), $\emptyset$, and $\mathbb{R}^n$.



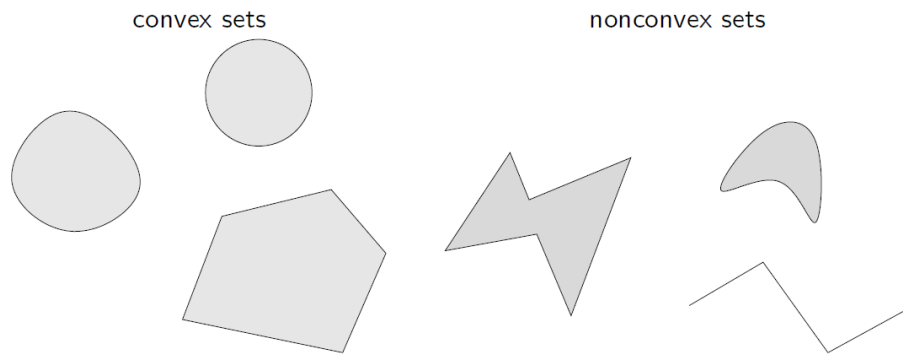


Figure 1: Examples of convex and nonconvex sets.

- A hyperplane is a set of the form

$$H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{x} = b\} \quad (\mathbf{a} \in \mathbb{R}^n \setminus \{\mathbf{0}\}, b \in \mathbb{R}) .$$

- The associated half-space is the set

$$H^- = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{x} \leq b\} .$$

Both hyperplanes and half-spaces are convex sets.

- The open ball

$$B(\mathbf{c}, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{c}\| < r\} ,$$

and the closed ball

$$B[\mathbf{c}, r] = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{c}\| \leq r\} ,$$

are convex. Note that the norm is an arbitrary norm defined over $\mathbb{R}^n$.

A relevant result in optimization is the convexity of ellipsoids. An ellipsoid is a set of the form

$$E = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x}^\top \mathbf{Q} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + c \leq 0\} ,$$

where $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is positive semidefinite, $\mathbf{b} \in \mathbb{R}^n$ and $c \in \mathbb{R}$.

Lemma. E is convex.

Proof. Write E as $E = \{\mathbf{x} \in \mathbb{R}^n : f(\mathbf{x}) \leq 0\}$ where $f(\mathbf{x}) \equiv \mathbf{x}^\top \mathbf{Q} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + c$. Then, take $\mathbf{x}, \mathbf{y} \in E$ and $\lambda \in [0, 1]$, and $f(\mathbf{x}) \leq 0, f(\mathbf{y}) \leq 0$. The vector $\mathbf{z} = \lambda \mathbf{x} + (1 - \lambda)\mathbf{y}$ satisfies

$$\mathbf{z}^\top \mathbf{Q} \mathbf{z} = \lambda^2 \mathbf{x}^\top \mathbf{Q} \mathbf{x} + (1 - \lambda)^2 \mathbf{y}^\top \mathbf{Q} \mathbf{y} + 2\lambda(1 - \lambda)\mathbf{x}^\top \mathbf{Q} \mathbf{y} ,$$



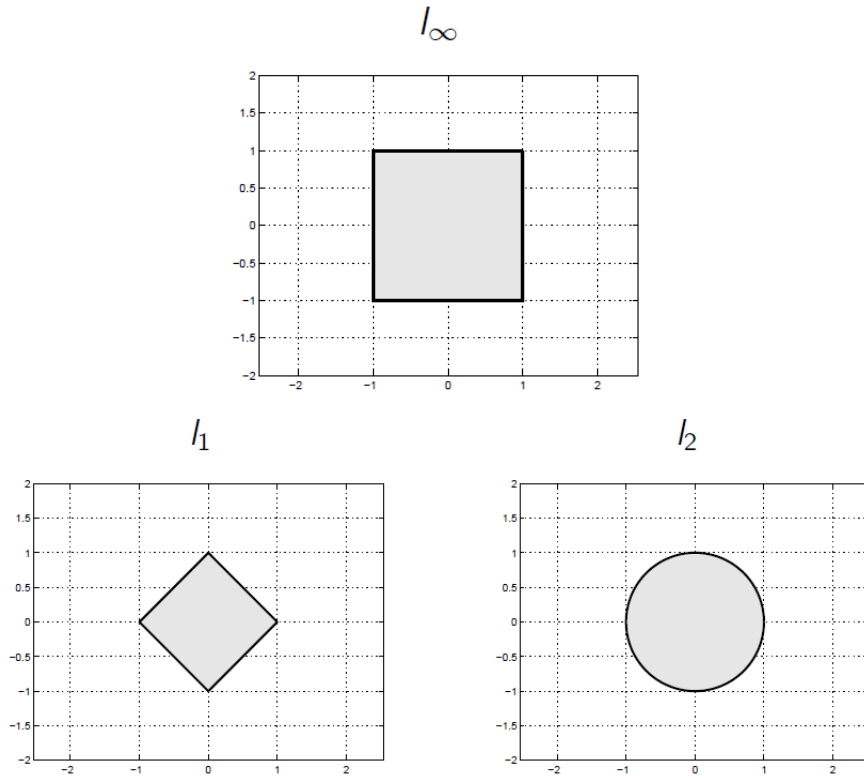


Figure 2: Different unit balls $\|\mathbf{x}\|_p \leq 1$ in $\mathbb{R}^2$.

and using Cauchy-Schwartz it follows that

$$\begin{aligned} \mathbf{x}^\top \mathbf{Q} \mathbf{y} &\leq \left\| \mathbf{Q}^{1/2} \mathbf{x} \right\| \cdot \left\| \mathbf{Q}^{1/2} \mathbf{y} \right\| = \sqrt{\mathbf{x}^\top \mathbf{Q} \mathbf{x}} \sqrt{\mathbf{y}^\top \mathbf{Q} \mathbf{y}} \\ &\leq \frac{1}{2} (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{y}^\top \mathbf{Q} \mathbf{y}) , \end{aligned}$$

which implies that

$$\mathbf{z}^\top \mathbf{Q} \mathbf{z} \leq \lambda \mathbf{x}^\top \mathbf{Q} \mathbf{x} + (1 - \lambda) \mathbf{y}^\top \mathbf{Q} \mathbf{y} .$$

Finally,

$$\begin{aligned} f(\mathbf{z}) &= \mathbf{z}^\top \mathbf{Q} \mathbf{z} + 2\mathbf{b}^\top \mathbf{z} + c \\ &\leq \lambda \mathbf{x}^\top \mathbf{Q} \mathbf{x} + (1 - \lambda) \mathbf{y}^\top \mathbf{Q} \mathbf{y} + 2\lambda \mathbf{b}^\top \mathbf{x} + 2(1 - \lambda) \mathbf{b}^\top \mathbf{y} + c \\ &= \lambda (\mathbf{x}^\top \mathbf{Q} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + c) + (1 - \lambda) (\mathbf{y}^\top \mathbf{Q} \mathbf{y} + 2\mathbf{b}^\top \mathbf{y} + c) \\ &= \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}) \leq 0 \end{aligned}$$

establishing the desired result that $\mathbf{z} \in E$. □

Algebraic Operations with Convex Sets

Lemma (Intersection of convex sets is convex). *Let $C_i \subseteq \mathbb{R}^n$ be a convex set for any $i \in I$ where I is an index set (possibly infinite). Then the set $\bigcap_{i \in I} C_i$ is convex.*



A direct consequence of the above is that convex polytopes of the form

$$P = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} \leq \mathbf{b}\},$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$ are convex since they are generated as the intersection of m half-spaces $\mathbf{a}_i^\top \mathbf{x} \leq b_i$.

Some important algebraic properties of convex sets are summarized in the following result:

Theorem. 1. Let $C_1, C_2, \dots, C_k \subseteq \mathbb{R}^n$ be convex sets and let $\mu_1, \mu_2, \dots, \mu_k \in \mathbb{R}$. Then the set $\mu_1 C_1 + \mu_2 C_2 + \dots + \mu_k C_k$ is convex.

2. Let $C_i \subseteq \mathbb{R}^{k_i}, i = 1, \dots, m$ be convex sets. Then the cartesian product

$$C_1 \times C_2 \times \dots \times C_m = \{(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m) : \mathbf{x}_i \in C_i, i = 1, 2, \dots, m\}$$

is convex.

3. Let $M \subseteq \mathbb{R}^n$ be a convex set and let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Then the set

$$\mathbf{A}(M) = \{\mathbf{Ax} : \mathbf{x} \in M\}$$

is convex.

4. Let $D \subseteq \mathbb{R}^m$ be convex and let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Then the set

$$\mathbf{A}^{-1}(D) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} \in D\}$$

is convex.

The Convex Hull

Definition (Convex Combinations). Given m points $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \in \mathbb{R}^n$, a convex combination of these m points is a vector of the form $\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2 + \dots + \lambda_m \mathbf{x}_m$, where $\lambda_1, \lambda_2, \dots, \lambda_m$ are nonnegative numbers satisfying $\lambda_1 + \lambda_2 + \dots + \lambda_m = 1$.

A convex set is defined by the property that any convex combination of two points from the set is also in the set. We will now show that a convex combination of any number of points from a convex set is in the set.

Theorem. Let $C \subseteq \mathbb{R}^n$ be a convex set and let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \in C$. Then for any $\boldsymbol{\lambda} \in \Delta_m$, the relation $\sum_{i=1}^m \lambda_i \mathbf{x}_i \in C$ holds.



Proof. Proof by induction on m . For $m = 1$ the result is obvious. The induction hypothesis is that for any m vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m \in C$ and any $\boldsymbol{\lambda} \in \Delta_m$, the vector $\sum_{i=1}^m \lambda_i \mathbf{x}_i$ belongs to C . We will now prove the theorem for $m + 1$ vectors. Suppose that $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{m+1} \in C$ and that $\boldsymbol{\lambda} \in \Delta_{m+1}$. We will show that $\mathbf{z} \equiv \sum_{i=1}^{m+1} \lambda_i \mathbf{x}_i \in C$. For this, if $\lambda_{m+1} = 1$, then $\mathbf{z} = \mathbf{x}_{m+1} \in C$ and the result obviously follows. Otherwise, if $\lambda_{m+1} < 1$, then

$$\begin{aligned} \mathbf{z} &= \sum_{i=1}^m \lambda_i \mathbf{x}_i + \lambda_{m+1} \mathbf{x}_{m+1} \\ &= (1 - \lambda_{m+1}) \sum_{i=1}^m \frac{\lambda_i}{1 - \lambda_{m+1}} \mathbf{x}_i + \lambda_{m+1} \mathbf{x}_{m+1}. \end{aligned}$$

Since $\sum_{i=1}^m \frac{\lambda_i}{1 - \lambda_{m+1}} = \frac{1 - \lambda_{m+1}}{1 - \lambda_{m+1}} = 1$, it follows that $\mathbf{v} = \sum_{i=1}^m \frac{\lambda_i}{1 - \lambda_{m+1}} \mathbf{x}_i$ is a convex combination of m points from C , and hence by the induction hypotheses we have that $\mathbf{v} \in C$. Thus, by the definition of a convex set, $\mathbf{z} = (1 - \lambda_{m+1}) \mathbf{v} + \lambda_{m+1} \mathbf{x}_{m+1} \in C$. $\square$

Definition (The Convex Hull). Let $S \subseteq \mathbb{R}^n$. The convex hull of S , denoted by $\text{conv}(S)$, is the set comprising all the convex combinations of vectors from S :

$$\text{conv}(S) \equiv \left\{ \sum_{i=1}^k \lambda_i \mathbf{x}_i : \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k \in S, \boldsymbol{\lambda} \in \Delta_k \right\}$$

The convex hull $\text{conv}(S)$ is "smallest" convex set containing S , in the sense that if another convex set T contains S , then $\text{conv}(S) \subset T$.

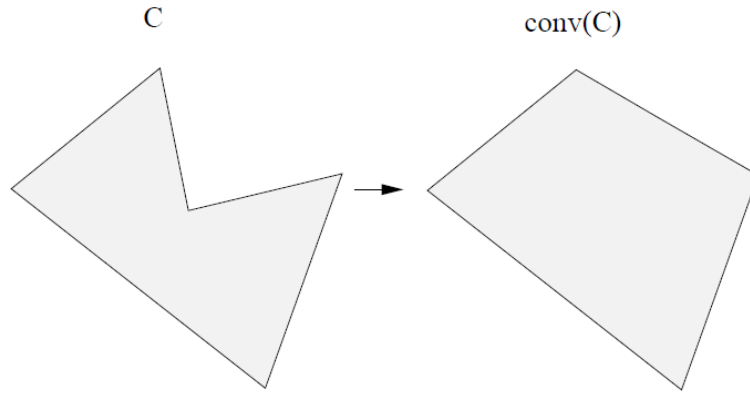


Figure 3: The convex hull of a non-convex set.

The following well-known result, called the Carathéodory theorem, states that any element in the convex hull of a subset of a given set $S \subset \mathbb{R}^n$ can be expressed as a **convex combination** of no more than $n + 1$ vectors from S .

Theorem (Carathéodory). Let $S \subseteq \mathbb{R}^n$ and let $\mathbf{x} \in \text{conv}(S)$. Then, there exist $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n+1} \in S$ such that $\mathbf{x} \in \text{conv}(\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n+1}\})$, that is, there exist $\boldsymbol{\lambda} \in \Delta_{n+1}$ such that

$$\mathbf{x} = \sum_{i=1}^{n+1} \lambda_i \mathbf{x}_i$$



We present this proof as it provides a construction mechanism.

Proof. Let $\mathbf{x} \in \text{conv}(S)$. Then, there exist vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k \in S$ and $\lambda \in \Delta_k$ such that

$$\mathbf{x} = \sum_{i=1}^k \lambda_i \mathbf{x}_i.$$

We can assume that $\lambda_i > 0$ for all $i = 1, 2, \dots, k$. If $k \leq n+1$, the result is proven. Otherwise, if $k \geq n+2$, then the vectors $\mathbf{x}_2 - \mathbf{x}_1, \mathbf{x}_3 - \mathbf{x}_1, \dots, \mathbf{x}_k - \mathbf{x}_1$, being more than n vectors in $\mathbb{R}^n$, are necessarily linearly dependent implying that there exist $\mu_2, \mu_3, \dots, \mu_k$ not all zeros such that

$$\sum_{i=2}^k \mu_i (\mathbf{x}_i - \mathbf{x}_1) = \mathbf{0}.$$

Defining $\mu_1 = -\sum_{i=2}^k \mu_i$, we obtain that

$$\sum_{i=1}^k \mu_i \mathbf{x}_i = \mathbf{0}.$$

Note that not all of the coefficients $\mu_1, \mu_2, \dots, \mu_k$ are zeros and $\sum_{i=1}^k \mu_i = 0$. There exists an index i for which $\mu_i < 0$. Let $\alpha \in \mathbb{R}_+$. Then,

$$\mathbf{x} = \sum_{i=1}^k \lambda_i \mathbf{x}_i = \sum_{i=1}^k \lambda_i \mathbf{x}_i + \alpha \sum_{i=1}^k \mu_i \mathbf{x}_i = \sum_{i=1}^k (\lambda_i + \alpha \mu_i) \mathbf{x}_i.$$

We have $\sum_{i=1}^k (\lambda_i + \alpha \mu_i) = 1$, so the equation above is a convex combination representation if and only if

$$\lambda_i + \alpha \mu_i \geq 0 \text{ for all } i = 1, \dots, k$$

but since $\lambda_i > 0$ for all i , it follows that these inequalities are satisfied for all $\alpha \in [0, \varepsilon]$ where $\varepsilon = \min_{i: \mu_i < 0} \left\{ -\frac{\lambda_i}{\mu_i} \right\}$. If we substitute $\alpha = \varepsilon$, then the inequalities still hold, but $\lambda_j + \varepsilon \mu_j = 0$ for $j \in \underset{i: \mu_i < 0}{\text{argmin}} \left\{ -\frac{\lambda_i}{\mu_i} \right\}$. This means that we found a representation of $\mathbf{x}$ as a convex combination of $k-1$ (or less) vectors. This process can be carried on until a representation of $\mathbf{x}$ as a convex combination of no more than $n+1$ vectors is derived. $\square$

Example For $n = 2$, consider the four vectors

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{x}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{x}_3 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \mathbf{x}_4 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

and let $\mathbf{x} \in \text{conv}(\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\})$ be given by

$$\mathbf{x} = \frac{1}{8}\mathbf{x}_1 + \frac{1}{4}\mathbf{x}_2 + \frac{1}{2}\mathbf{x}_3 + \frac{1}{8}\mathbf{x}_4 = \begin{pmatrix} \frac{13}{8} \\ \frac{11}{8} \end{pmatrix}$$

Find a representation of $\mathbf{x}$ as a convex combination of no more than 3 vectors.



Definition (Extreme Point). Let $S \subseteq \mathbb{R}^n$ be a convex set. A point $\mathbf{x} \in S$ is called an extreme point of S if there do not exist $\mathbf{x}_1, \mathbf{x}_2 \in S$ ($\mathbf{x}_1 \neq \mathbf{x}_2$) and $\lambda \in (0, 1)$, such that $\mathbf{x} = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2$. The set of extreme point is denoted by $\text{ext}(S)$.

For example, the set of extreme points of a convex polytope consists of all its vertices.

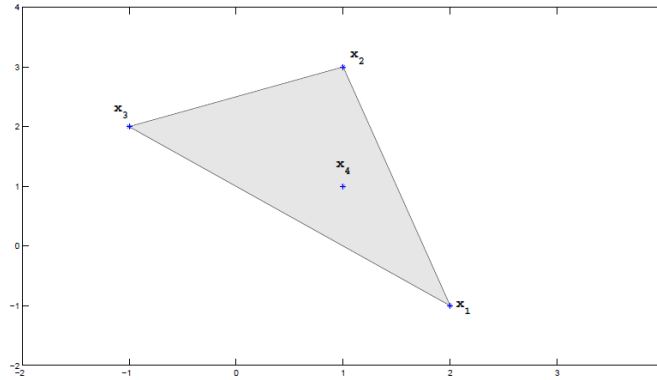


Figure 4: The extreme points of this triangle are given by its vertices $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$. The point $\mathbf{x}_4$ is not an extreme point.

Theorem (The Krein-Milman Theorem). Let $S \subseteq \mathbb{R}^n$ be a compact convex set. Then

$$S = \text{conv}(\text{ext}(S)) .$$

Convex Functions

We begin by giving a definition of convex function.

Definition (Convex Function). A function $f : C \rightarrow \mathbb{R}$ defined on a convex set $C \subseteq \mathbb{R}^n$ is called convex (or convex over C) if

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}) \text{ for any } \mathbf{x}, \mathbf{y} \in C, \lambda \in [0, 1]$$

Convexity, Strict Convexity and Concavity

In case where no domain is specified, we naturally assume that f is defined over the entire space $\mathbb{R}^n$.

Definition (Strict Convexity). A function $f : C \rightarrow \mathbb{R}$ defined on a convex set $C \subseteq \mathbb{R}^n$ is called strictly convex if

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) < \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}) \text{ for any } \mathbf{x} \neq \mathbf{y} \in C, \lambda \in (0, 1)$$



Definition (Concavity). A function is called concave if $-f$ is convex. Similarly, f is called strictly concave if $-f$ is strictly convex. We can also define concavity directly: a function f is concave if and only if for any $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in [0, 1]$

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \geq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y})$$

Examples of Convex Functions

- Affine Functions. $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$, where $\mathbf{a} \in \mathbb{R}^n$ and $b \in \mathbb{R}$. Take $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\lambda \in [0, 1]$. Then

$$\begin{aligned} f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) &= \mathbf{a}^\top (\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) + b \\ &= \lambda (\mathbf{a}^\top \mathbf{x}) + (1 - \lambda) (\mathbf{a}^\top \mathbf{y}) + \lambda b + (1 - \lambda) b \\ &= \lambda (\mathbf{a}^\top \mathbf{x} + b) + (1 - \lambda) (\mathbf{a}^\top \mathbf{y} + b) \\ &= \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}) . \end{aligned}$$

- Norms. $g(\mathbf{x}) = \|\mathbf{x}\|$, take $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\lambda \in [0, 1]$. Then

$$\begin{aligned} g(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) &= \|\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}\| \\ &\leq \|\lambda \mathbf{x}\| + \|(1 - \lambda) \mathbf{y}\| \\ &= \lambda \|\mathbf{x}\| + (1 - \lambda) \|\mathbf{y}\| \\ &= \lambda g(\mathbf{x}) + (1 - \lambda) g(\mathbf{y}) . \end{aligned}$$

We now state a fundamental result for convex functions.

Theorem (Jensen's Inequality). Let $f : C \rightarrow \mathbb{R}$ be a convex function where $C \subseteq \mathbb{R}^n$ is a convex set. Then, for any $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k \in C$ and $\lambda \in \Delta_k$, the following inequality holds:

$$f\left(\sum_{i=1}^k \lambda_i \mathbf{x}_i\right) \leq \sum_{i=1}^k \lambda_i f(\mathbf{x}_i) .$$

An important set associated with convex functions is the *epigraph*.

Definition (epigraph). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Then the epigraph set $\text{epi}(f) \subseteq \mathbb{R}^{n+1}$ is defined by

$$\text{epi}(f) = \left\{ \begin{pmatrix} \mathbf{x} \\ t \end{pmatrix} : f(\mathbf{x}) \leq t \right\} .$$

An example of an epigraph is shown in Figure 5.

The definition of the epigraph allows us to connect the notions of convex sets and convex functions through the following theorem.

Theorem. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Then the function f is convex if and only if its epigraph is a convex set.



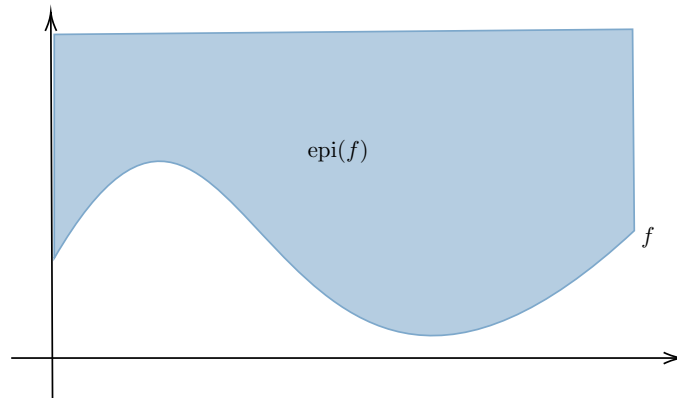


Figure 5: The epigraph of a one-dimensional function.

First-order Characterization of Convex Functions

Convex functions are not necessarily differentiable (think about $f(x) = |x|$), but in case they are, we can generalize first and a second-order optimality conditions for functions such as those from Week 3.

Theorem (The Gradient Inequality). *Let $f : C \rightarrow \mathbb{R}$ be a continuously differentiable function defined on a convex set $C \subseteq \mathbb{R}^n$. Then f is convex over C if and only if*

$$f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) \text{ for any } \mathbf{x}, \mathbf{y} \in C.$$

An analogous result holds for strictly convex functions (with a strict inequality).

Proof. Suppose first that f is convex. Let $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in (0, 1]$. If $\mathbf{x} = \mathbf{y}$, then the inequality trivially holds. We will therefore assume that $\mathbf{x} \neq \mathbf{y}$. The

$$\frac{f(\mathbf{x} + \lambda(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\lambda} \leq f(\mathbf{y}) - f(\mathbf{x}).$$

Taking $\lambda \rightarrow 0^+$, we obtain

$$f'(\mathbf{x}; \mathbf{y} - \mathbf{x}) \leq f(\mathbf{y}) - f(\mathbf{x}).$$

Since f is continuously differentiable, $f'(\mathbf{x}; \mathbf{y} - \mathbf{x}) = \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$, and the inequality follows.



To prove the reverse direction, assume that the gradient inequality holds. Let $\mathbf{z}, \mathbf{w} \in C$, and let $\lambda \in (0, 1)$. We will show that $f(\lambda\mathbf{z} + (1 - \lambda)\mathbf{w}) \leq \lambda f(\mathbf{z}) + (1 - \lambda)f(\mathbf{w})$. Let $\mathbf{u} = \lambda\mathbf{z} + (1 - \lambda)\mathbf{w} \in C$. Then

$$\mathbf{z} - \mathbf{u} = \frac{\mathbf{u} - (1 - \lambda)\mathbf{w}}{\lambda} - \mathbf{u} = -\frac{1 - \lambda}{\lambda}(\mathbf{w} - \mathbf{u}).$$

We have

$$\begin{aligned} f(\mathbf{u}) + \nabla f(\mathbf{u})^\top (\mathbf{z} - \mathbf{u}) &\leq f(\mathbf{z}), \\ f(\mathbf{u}) - \frac{\lambda}{1 - \lambda} \nabla f(\mathbf{u})^\top (\mathbf{z} - \mathbf{u}) &\leq f(\mathbf{w}). \end{aligned}$$

Thus,

$$f(\mathbf{u}) \leq \lambda f(\mathbf{z}) + (1 - \lambda)f(\mathbf{w})$$

□

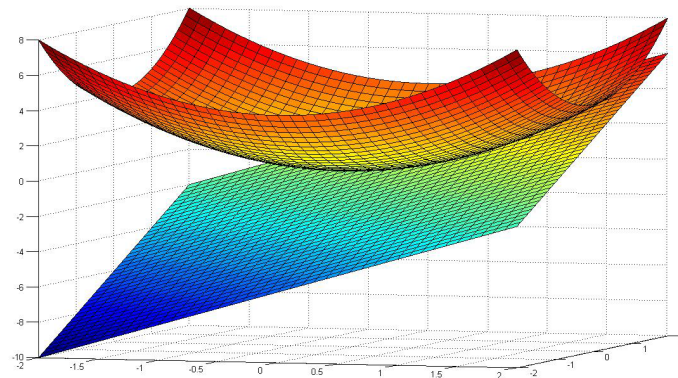


Figure 6: For a convex function f , the tangent plane at every point is always below f .

Convexity + Stationarity $\Rightarrow$ Global Optimality! A direct result of the gradient inequality is that the first order optimality condition $\nabla f(\mathbf{x}^*) = \mathbf{0}$ is sufficient for global optimality.

Theorem (Stationarity Implies Global Optimality). *Let f be a continuously differentiable function which is convex over a convex set $C \subseteq \mathbb{R}^n$. Suppose that $\nabla f(\mathbf{x}^*) = \mathbf{0}$ for some $\mathbf{x}^* \in C$. Then $\mathbf{x}^*$ is the global minimizer of f over C .*

We now revisit optimality conditions for quadratic functions.

Theorem (Convexity of Quadratic Functions with Positive Semidefinite Matrices). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be the quadratic function given by $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + c$ where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, $\mathbf{b} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. Then f is (strictly) convex if and only if $\mathbf{A} \geq \mathbf{0}$ ($\mathbf{A} > \mathbf{0}$).*



Proof. The convexity of f is equivalent to

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) \text{ for any } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

This is the same as stating that

$$\mathbf{y}^\top \mathbf{A} \mathbf{y} + 2\mathbf{b}^\top \mathbf{y} + c \geq \mathbf{x}^\top \mathbf{A} \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + c + 2(\mathbf{A} \mathbf{x} + \mathbf{b})^\top (\mathbf{y} - \mathbf{x}),$$

for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, from where it follows that

$$(\mathbf{y} - \mathbf{x})^\top \mathbf{A} (\mathbf{y} - \mathbf{x}) \geq 0,$$

for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. This is equivalent to the inequality $\mathbf{d}^\top \mathbf{A} \mathbf{d} \geq 0$ for any $\mathbf{d} \in \mathbb{R}^n$, which is the same as $\mathbf{A} \geq \mathbf{0}$. Similar arguments show that strict convexity is equivalent to

$$\mathbf{d}^\top \mathbf{A} \mathbf{d} > 0 \text{ for any } \mathbf{0} \neq \mathbf{d} \in \mathbb{R}^n,$$

namely to $\mathbf{A} > \mathbf{0}$. □

Theorem (Monotonicity of the Gradient). *Suppose that f is a continuously differentiable function over a convex set $C \subseteq \mathbb{R}^n$. Then f is convex over C if and only if*

$$(\nabla f(\mathbf{x}) - \nabla f(\mathbf{y}))^\top (\mathbf{x} - \mathbf{y}) \geq 0 \text{ for any } \mathbf{x}, \mathbf{y} \in C.$$

Second-order Characterization of Convex Functions

We can now extend our link between convexity and optimality conditions to second-order characterizations.

Theorem (Second-Order Characterization of Convexity). *Let f be a twice continuously differentiable function over an open convex set $C \subseteq \mathbb{R}^n$. Then f is convex over C if and only if $\nabla^2 f(\mathbf{x}) \geq \mathbf{0}$ for any $\mathbf{x} \in C$.*

Example. Convexity of the log-sum-exp function:

$$f(\mathbf{x}) = \log(e^{x_1} + e^{x_2} + \dots + e^{x_n}), \quad \mathbf{x} \in \mathbb{R}^n.$$

The gradient is given by:

$$\frac{\partial f}{\partial x_i}(\mathbf{x}) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}, \quad i = 1, 2, \dots, n.$$

Therefore, the Hessian is computed as

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}) = \begin{cases} -\frac{e^{x_i} e^{x_j}}{(\sum_{j=1}^n e^{x_j})^2}, & i \neq j \\ -\frac{e^{x_i} e^{x_j}}{(\sum_{j=1}^n e^{x_j})^2} + \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}, & i = j \end{cases}.$$



We can thus write the Hessian matrix as

$$\nabla^2 f(\mathbf{x}) = \text{diag}(\mathbf{w}) - \mathbf{w}\mathbf{w}^\top, \quad \text{with} \quad \mathbf{w} = \left(\frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}} \right)_{i=1}^n \in \Delta_n.$$

For any $\mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v} = \sum_{i=1}^n w_i v_i^2 - (\mathbf{v}^\top \mathbf{w})^2 \geq 0,$$

since defining $s_i = \sqrt{w_i} v_i$, $t_i = \sqrt{w_i}$, we have

$$(\mathbf{v}^\top \mathbf{w})^2 = (\mathbf{s}^\top \mathbf{t})^2 \leq \|\mathbf{s}\|^2 \|\mathbf{t}\|^2 = \left(\sum_{i=1}^n w_i v_i^2 \right) \left(\sum_{i=1}^n w_i \right) = \sum_{i=1}^n w_i v_i^2.$$

Thus, $\nabla^2 f(\mathbf{x}) \geq \mathbf{0}$ and hence f is convex over $\mathbb{R}^n$.

Example Show the convexity of the quad-over-lin function

$$f(x_1, x_2) = \frac{x_1^2}{x_2}$$

defined over $\mathbb{R} \times \mathbb{R}_{++} = \{(x_1, x_2) : x_2 > 0\}$.

Further Results for Convex Functions

Operations Preserving Convexity

- Let f be a convex function defined over a convex set $C \subseteq \mathbb{R}^n$ and let $\alpha \geq 0$. Then αf is a convex function over C .
- Let $f_1, f_2, \dots, f_p$ be convex functions over a convex set $C \subseteq \mathbb{R}^n$. Then the sum function $f_1 + f_2 + \dots + f_p$ is convex over C .
- Let f be a convex function defined on a convex set $C \subseteq \mathbb{R}^n$. Let $\mathbf{A} \in \mathbb{R}^{n \times m}$ and $\mathbf{b} \in \mathbb{R}^m$. Then the function g defined by

$$g(\mathbf{y}) = f(\mathbf{A}\mathbf{y} + \mathbf{b})$$

is convex over the convex set $D = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{A}\mathbf{y} + \mathbf{b} \in C\}$.

- Let $f : C \rightarrow \mathbb{R}$ be a convex function defined over the convex set $C \subseteq \mathbb{R}^n$. Let $g : I \rightarrow \mathbb{R}$ be a one-dimensional nondecreasing convex function over the interval $I \subseteq \mathbb{R}$. Assume that the image of C under f is contained in I : $f(C) \subseteq I$. Then the composition of g with f defined by

$$h(\mathbf{x}) \equiv g(f(\mathbf{x}))$$

is convex over C .



Several Examples of Convex Functions Using these Properties

- The generalized quad-over-lin function

$$g(\mathbf{x}) = \frac{\|\mathbf{Ax} + \mathbf{b}\|^2}{\mathbf{c}^\top \mathbf{x} + d} \quad (\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n, d \in \mathbb{R})$$

is convex over $D = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{c}^\top \mathbf{x} + d > 0\}$.

- $f(x_1, x_2) = -\log(x_1 x_2)$, over $\mathbb{R}_{++}^2$.
- $f(x_1, x_2) = x_1^2 + 2x_1 x_2 + 3x_2^2 + 2x_1 - 3x_2 + e^{x_1}$.
- $h(\mathbf{x}) = e^{\|\mathbf{x}\|^2}$.

Theorem (Point-Wise Maximum of Convex Functions). *Let $f_1, f_2, \dots, f_p : C \rightarrow \mathbb{R}$ be p convex functions over the convex set $C \subseteq \mathbb{R}^n$. Then the maximum function*

$$f(\mathbf{x}) \equiv \max_{i=1,2,\dots,p} \{f_i(\mathbf{x})\}$$

is convex over C .

Examples

- $f(\mathbf{x}) = \max \{x_1, x_2, \dots, x_n\}$ is convex.
- For a given vector $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top \in \mathbb{R}^n$, let $x_{[i]}$ denote the i -th largest value in $\mathbf{x}$. For any $k \in \{1, 2, \dots, n\}$ the function

$$h_k(\mathbf{x}) = x_{[1]} + x_{[2]} + \dots + x_{[k]}$$

is convex.

Theorem (Preservation of Convexity Under Partial Minimization). *Let $f : C \times D \rightarrow \mathbb{R}$ be a convex function defined over the set $C \times D$ where $C \subseteq \mathbb{R}^m$ and $D \subseteq \mathbb{R}^n$ are convex sets. Let*

$$g(\mathbf{x}) = \min_{\mathbf{y} \in D} f(\mathbf{x}, \mathbf{y}), \quad \mathbf{x} \in C$$

where we assume that the minimum is finite. Then g is convex over C .

Example. The distance function from a convex set $d_C(\mathbf{x}) \equiv \inf_{\mathbf{y} \in C} \|\mathbf{x} - \mathbf{y}\|$ is convex.



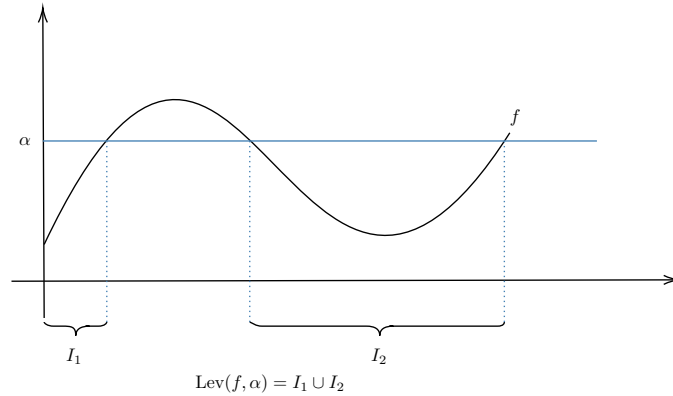


Figure 7: Example of a level set of a one-dimensional function f at level α .

Level Sets of Convex Functions

Let us start with the definition of a *level set*.

Definition (level sets). *Let $f: S \rightarrow \mathbb{R}$ be a function defined over a set $S \subseteq \mathbb{R}^n$. Then the level set of f with level α is given by*

$$\text{Lev}(f, \alpha) = \{\mathbf{x} \in S: f(\mathbf{x}) \leq \alpha\}.$$

An example of a level set of a one-dimensional function f at level α is shown in Figure 7.

In what follows, we present a fundamental property of convex functions with respect to their level sets.

Theorem (convexity of level sets of convex functions). *Let $f: C \rightarrow \mathbb{R}$ be a convex function defined over a convex set $C \subseteq \mathbb{R}^n$. Then for any $\alpha \in \mathbb{R}$ the level set $\text{Lev}(f, \alpha)$ is convex.*

The previous theorem states that all convex functions have all their level sets convex. However, the opposite is not true. In fact, there exists nonconvex functions whose level sets are all convex: for example, consider the function $f(\mathbf{x}) = \sqrt{|\mathbf{x}|}$.

Four Important Theorems for Convex Functions

We now state four important results for convex functions.

Theorem (Continuity of Convex Functions). *Let $f : C \rightarrow \mathbb{R}$ be a convex function defined over a convex set $C \subseteq \mathbb{R}^n$. Let $\mathbf{x}_0 \in \text{int}(C)$. Then there exist $\varepsilon > 0$ and $L > 0$ such that $B[\mathbf{x}_0, \varepsilon] \subseteq C$ and*

$$|f(\mathbf{x}) - f(\mathbf{x}_0)| \leq L \|\mathbf{x} - \mathbf{x}_0\| \text{ for any } \mathbf{x} \in B[\mathbf{x}_0, \varepsilon]$$

Theorem (Existence of Directional Derivatives of Convex Functions). *Let $f : C \rightarrow \mathbb{R}$ be a convex function over the convex set $C \subseteq \mathbb{R}^n$. Let $\mathbf{x} \in \text{int}(C)$. Then for any $\mathbf{d} \neq 0$, the directional derivative $f'(\mathbf{x}; \mathbf{d})$ exists.*

The last two theorems relate to the problem of **maximizing** a non-constant convex function over a convex set.

Theorem (No Maximum Inside the Convex Set). *Let $f : C \rightarrow \mathbb{R}$ be convex and non-constant over the nonempty convex set $C \subseteq \mathbb{R}^n$. Then f does not attain a maximum at a point in $\text{int}(C)$.*

Finally, we state that maximum of convex function over compact convex sets can be found at the extreme points of the set.

Theorem (Maximum of a Convex Function Over a Compact Convex Set). *Let $f : C \rightarrow \mathbb{R}$ be convex over the nonempty convex and compact set $C \subseteq \mathbb{R}^n$. Then there exists at least one maximizer of f over C that is an extreme point of C .*

Proof. Let $\mathbf{x}^*$ be a maximizer of f over C . If $\mathbf{x}^*$ is an extreme point of C , then the result is established. Otherwise, $\mathbf{x}^*$ is not an extreme point, by Krein-Milman, $C = \text{conv}(\text{ext}(C))$ implies the existence of $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k \in \text{ext}(C)$ such that

$$\mathbf{x}^* = \sum_{i=1}^k \lambda_i \mathbf{x}_i$$

By the convexity of f ,

$$f(\mathbf{x}^*) \leq \sum_{i=1}^k \lambda_i f(\mathbf{x}_i)$$

or equivalently

$$\sum_{i=1}^k \lambda_i (f(\mathbf{x}_i) - f(\mathbf{x}^*)) \geq 0.$$

Since $\mathbf{x}^*$ is a maximizer of f over C , we have $f(\mathbf{x}_i) \leq f(\mathbf{x}^*)$ for all $i = 1, \dots, k$. This implies that $f(\mathbf{x}_i) = f(\mathbf{x}^*)$. Consequently, the extreme points $\mathbf{x}_1, \dots, \mathbf{x}_k$ are all maximizers of f over C . $\square$



Convexity and Inequalities

In what follows we will show that the notions of convexity allows us to prove well-known inequalities. For example, we focus on the Arithmetic-Geometric Mean inequality, Young's Hölder's and Minkowski's inequalities.

Lemma (Arithmetic-Geometric Mean). *For all $\mathbf{x} \in \mathbb{R}_+^n$, then*

$$\sum_{i=1}^n \lambda_i x_i \geq \prod_{i=1}^n x_i^{\lambda_i}$$

for $\lambda \in \Delta_n$.

Proof. We start by noticing that if $\mathbf{x} \in \mathbb{R}^n$ has at least one entry equal to 0, then the inequality is satisfied trivially. For this reason, we can assume $\mathbf{x} \in \mathbb{R}_{++}^n$. Let us consider the function $f(t) = \log(t)$ for which we have already proved its convexity for all $t > 0$. Using Jensen's inequality, it holds

$$\begin{aligned} f\left(\sum_{i=1}^n \lambda_i x_i\right) &\leq \sum_{i=1}^n \lambda_i f(x_i) \\ -\log\left(\sum_{i=1}^n \lambda_i x_i\right) &\leq -\sum_{i=1}^n \lambda_i \log(x_i) \end{aligned}$$

Now, using the properties of the log, we obtain

$$\begin{aligned} \log\left(\sum_{i=1}^n \lambda_i x_i\right) &\geq \sum_{i=1}^n \lambda_i \log(x_i^{\lambda_i}) \\ \sum_{i=1}^n \lambda_i x_i &\geq \prod_{i=1}^n x_i^{\lambda_i} \end{aligned} \quad \square$$

The next inequality is obtained as a direct result of the AGM inequality

Lemma (Young's inequality). *For any $s, t \geq 0$ and $p, q > 1$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$ it holds that*

$$st \leq \frac{s^p}{p} + \frac{t^q}{q}.$$

Proof. Using the AGM-inequality, we have that for $x, y \geq 0$

$$x^{1/p} y^{1/q} \leq \frac{x}{p} + \frac{y}{q}.$$

Now, choosing $x = s^p$ and $y = t^q$, we obtain the result. $\square$

Now, we focus on the generalization of the Cauchy-Schwarz inequality, called Hölder's inequality.



Lemma (Hölder's inequality). For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $p, q \geq 1$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$ it holds that

$$|\mathbf{x}^\top \mathbf{y}| \leq \|\mathbf{x}\|_p \|\mathbf{y}\|_q.$$

Proof. Let us consider $\mathbf{x} \neq 0, \mathbf{y} \neq 0$, since for $\mathbf{x} = 0$ or $\mathbf{y} = 0$ the inequality is trivially satisfied. Now, using Young's inequality with $s = \frac{|x_i|}{\|\mathbf{x}\|_p}$ and $t = \frac{|y_i|}{\|\mathbf{y}\|_q}$ we obtain

$$\frac{|x_i y_i|}{\|\mathbf{x}\|_p \|\mathbf{y}\|_q} \leq \frac{1}{p} \frac{|x_i|^p}{\|\mathbf{x}\|_p^p} + \frac{1}{q} \frac{|y_i|^q}{\|\mathbf{y}\|_q^q} \quad \text{for all } i = 1, \dots, n.$$

Summing over $i = 1, 2, \dots, n$ we obtain

$$\frac{1}{\|\mathbf{x}\|_p \|\mathbf{y}\|_q} \sum_{i=1}^n |x_i y_i| \leq \frac{1}{p \|\mathbf{x}\|_p^p} \sum_{i=1}^n |x_i|^p + \frac{1}{q \|\mathbf{y}\|_q^q} \sum_{i=1}^n |y_i|^q = \frac{1}{p} + \frac{1}{q} = 1.$$

Finally using triangular inequality we obtain

$$|\mathbf{x}^\top \mathbf{y}| \leq \sum_{i=1}^n |x_i y_i| \leq \|\mathbf{x}\|_p \|\mathbf{y}\|_q. \quad \square$$

We end by showing the Minkowski's inequality which shows the triangle inequality for p -norms with $p \geq 1$.

Lemma (Minkowski's inequality). Let $p \geq 1$. Then for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ the inequality

$$\|\mathbf{x} + \mathbf{y}\|_p \leq \|\mathbf{x}\|_p + \|\mathbf{y}\|_p;$$

holds.

Proof. For $p = 1$, the inequality follows automatically from the properties of the absolute value and then summing over $i = 1, \dots, n$. Thus, we focus on the case $p > 1$, we can assume moreover, that $\mathbf{x} \neq 0, \mathbf{y} \neq 0$ and $\mathbf{x} + \mathbf{y} \neq 0$, otherwise the inequality is trivially satisfied. Let us consider the function $\varphi: \mathbb{R}_{++} \rightarrow \mathbb{R}$ such that $\varphi(t) = t^p$, which we know is convex since $\varphi''(t) = p(p-1)t^{p-2} > 0$, for $t > 0$. Now, recalling the definition of convexity we have

$$(\lambda_1 t + \lambda_2 s)^p \leq \lambda_1 t^p + \lambda_2 s^p$$

with $\lambda_1, \lambda_2 \geq 0$ and $\lambda_1 + \lambda_2 = 1$. Choosing in particular $\lambda_1 = \frac{\|\mathbf{x}\|_p}{\|\mathbf{x}\|_p + \|\mathbf{y}\|_p}$, $\lambda_2 = \frac{\|\mathbf{y}\|_p}{\|\mathbf{x}\|_p + \|\mathbf{y}\|_p}$, $t = \frac{|x_i|}{\|\mathbf{x}\|_p}$ and $s = \frac{|y_i|}{\|\mathbf{y}\|_p}$ we obtain

$$\frac{1}{(\|\mathbf{x}\|_p + \|\mathbf{y}\|_p)^p} (|x_i| + |y_i|)^p \leq \frac{\|\mathbf{x}\|_p}{\|\mathbf{x}\|_p + \|\mathbf{y}\|_p} \frac{|x_i|^p}{\|\mathbf{x}\|_p^p} + \frac{\|\mathbf{y}\|_p}{\|\mathbf{x}\|_p + \|\mathbf{y}\|_p} \frac{|y_i|^p}{\|\mathbf{y}\|_p^p},$$

for all $i = 1, 2, \dots, n$. Finally, summing over $i = 1, \dots, n$ we obtain that

$$\frac{1}{(\|\mathbf{x}\|_p + \|\mathbf{y}\|_p)^p} \sum_{i=1}^n (|x_i| + |y_i|)^p \leq \frac{\|\mathbf{x}\|_p}{\|\mathbf{x}\|_p + \|\mathbf{y}\|_p} + \frac{\|\mathbf{y}\|_p}{\|\mathbf{x}\|_p + \|\mathbf{y}\|_p} = 1.$$

Finally,

$$\|\mathbf{x} + \mathbf{y}\|_p = \sqrt[p]{\sum_{i=1}^n |x_i + y_i|^p} \leq \sqrt[p]{\sum_{i=1}^n (|x_i| + |y_i|)^p} \leq \|\mathbf{x}\|_p + \|\mathbf{y}\|_p \quad \square$$

